ACCESS CONTROL & INTRUSION DETECTION

CYS335

INTRUSION DETECTION & PREVENTION SYSTEMS

CHAPTER ONE

INTRUSION DETECTION SYSTEMS

UNIT 1

HISTRORY OF INTRUSION DETECTION SYSTEMS

- System administrators performed intrusion detection by sitting in front of a console and monitoring user activities.
- Although effective enough at the time, this early form of intrusion detection was **ad hoc** and not scalable.
- The next step in intrusion detection involved **audit logs**, which system administrators reviewed for evidence of unusual or malicious behavior.
- As storage became cheaper, audit logs moved **online** and researchers developed programs to analyze the data.
- In the early '90s, researchers developed **real-time Intrusion Detection Systems** that reviewed audit data as it was produced.

INTRUSION DETECTION OVERVIEW

- A device that generates alerts when it observes potentially malicious traffic is called an **Intrusion Detection System (IDS)**.
- IDSs can be **software applications** that are installed on endpoints or dedicated **hardware** devices that are connected to the network.
- The goal of intrusion detection is seemingly simple, to **detect intrusions**.
- If there is no **manifestation** or if the information it contains is untrustworthy, then the system cannot detect the intrusion.

DETECTION TECHNIQUES

1. Anomaly Detection

- Creates a traffic **profile** as it observes traffic in normal operation.
- It then looks for packet streams that are statistically unusual.
- The great thing about anomaly-based IDS systems is that they **don't rely on previous knowledge** about existing attacks.
- Anomaly detection uses models of the intended behavior of users and applications, interpreting deviations from this "**normal**" behavior as a problem.
- A basic assumption of anomaly detection is that **attacks differ from normal behavior**.
- The main advantage of anomaly detection systems is that they **can detect previously unknown attacks.**
- In actual systems, however, the advantage of detecting previously unknown attacks is paid for in terms of **high false positive rates**.

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DETECTION TECHNIQUES (CONTINUES…)

2. Misuse Detection

- Essentially define what's wrong.
- They contain attack descriptions (or "**signatures**") and match them against the audit data stream, looking for evidence of known attacks.
- The **main advantage** of misuse detection systems is that they focus analysis on the audit data and typically **produce few false positives**.
- The **main disadvantage** of misuse detection systems is that they **can detect only known attacks** for which they have a defined signature.
- As new attacks are discovered, developers must model and add them to the signature database.

DETECTION TECHNIQUES (CONTINUES…)

3. Signature-based Detection

- Maintains a database of attack signatures against which it compares network packets.
- If a packet triggers a match to one of the signatures, the IDS flags it.
- In other words, a signature-based IDS is completely **blind to new attacks** that have yet to be recorded.

DETECTION TECHNIQUES (CONTINUES…)

4. Hybrid Detection Method

- Uses both **Signature** and **Anomaly**-based intrusion detection methods.
- However, the main reason behind the development of a hybrid detection system is to identify more potential attacks with **fewer errors**.

TOP INTRUSION DETECTION SYSTEM TOOLS

1. **McAfee**

- Is an IDS designed to identify **real-time** malicious activities.
- It uses both **signature** and **anomaly** methods to identify the threats with emulation techniques.
- Therefore, McAfee is a **scalable** application.

TOP INTRUSION DETECTION SYSTEM TOOLS (CONTINUES…)

2. Suricata

- It is an **open-source** tool based on a **Network Intrusion Detection System (NIDS)**.
- It detects identified threats and malicious activity in **real-time**.
- It uses a **Signature-based** method to identify the known threats or intrusion.

TOP INTRUSION DETECTION SYSTEM TOOLS (CONTINUES…)

3. Cisco Stealthwatch

- Cisco designed Stealthwatch with **NIDS** and **HIDS**.
- It is compatible with **Windows, Linux,** and **MacOS** operating systems.
- it uses **machine learning** to create baselines of patterns which is acceptable.
- One of the best features of this tool is that it can also detect intrusion and suspicious activities in the encrypted network without decrypting it.

RESPONSE: AFTER THE INTRUSION

- An intrusion detection system's **response** is its output or action upon detecting a problem.
- A response can take many different forms; the most common is to **generate an alert** that describes the detected intrusion.
- There are also more **aggressive responses**, such as **sounding a siren**, or even **mounting a counter-attack**.
- A **counterattack** might include reconfiguring a router to block the attacker's address or even attacking the culprit.
- Obviously, **aggressive responses can be dangerous**, since they could be launched against innocent victims.

TYPE OF THREATS

1. Internal Threats

- The term "**internal attack**" is used to describe an attack being implemented from a person or organization with some level of authorized access on your network.
- Internal attacks are performed from within the trusted area of the network.

TYPE OF THREATS (CONTINUES…)

2. External Threats

- Includes anyone that doesn't have authorized access to the internal network but attempts to gain access.
- External threats are what companies spend the most time and money trying to prevent.

NEED OF INTRUSION DETECTION SYSTEMS (IDS)

1. **Monitor** inbound and outbound network traffic.
2. **Analyze** the patterns in the network continuously.
3. Send an **alarm** immediately after detecting unwanted intrusion and activities in the network.

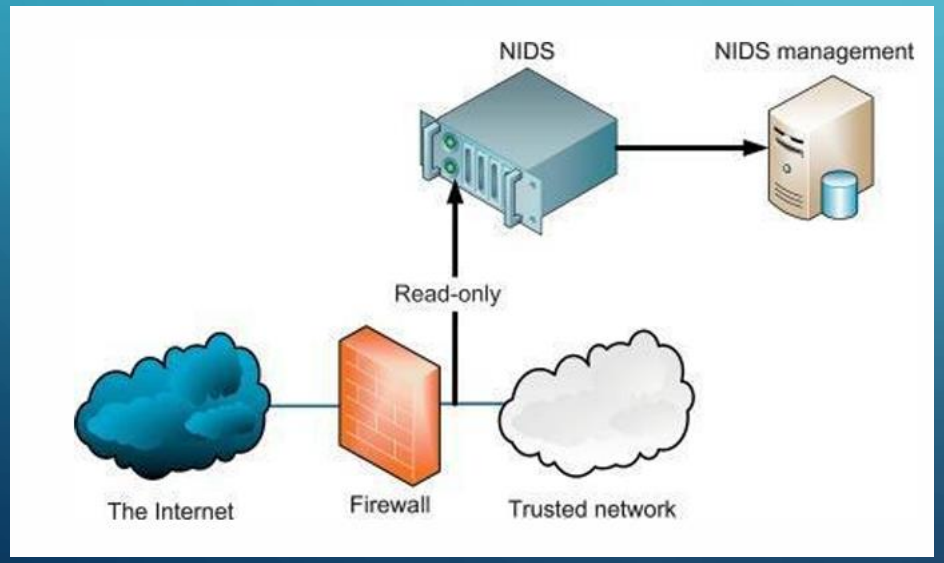
TYPES OF INTRUSION DETECTION SYSTEMS

1. **Network Intrusion Detection System (NIDS)**

- NIDS **monitors** the traffic on the network from all devices.
- Similarly, it examines the traffic passing on the entire subnet and verifies it with the packet metadata and content.
- If NIDS detects any intrusion in the network, a warning alert is sent to the admin of that network.
- Instead of analyzing information that originates and resides on a computer, network-based IDS uses techniques like "**packet-sniffing**" to pull data from TCP/IP or other protocol packets traveling along the network.

TYPES OF INTRUSION DETECTION SYSTEMS (CONTINUES…)

- NIDS usually **require promiscuous network** access in order to analyze all traffic, including all uncast traffic.
- The NIDS sniffs the internal interface of the firewall in **read-only** mode and sends alerts to a NIDS Management server via a different (i.e., read/write) network interface.



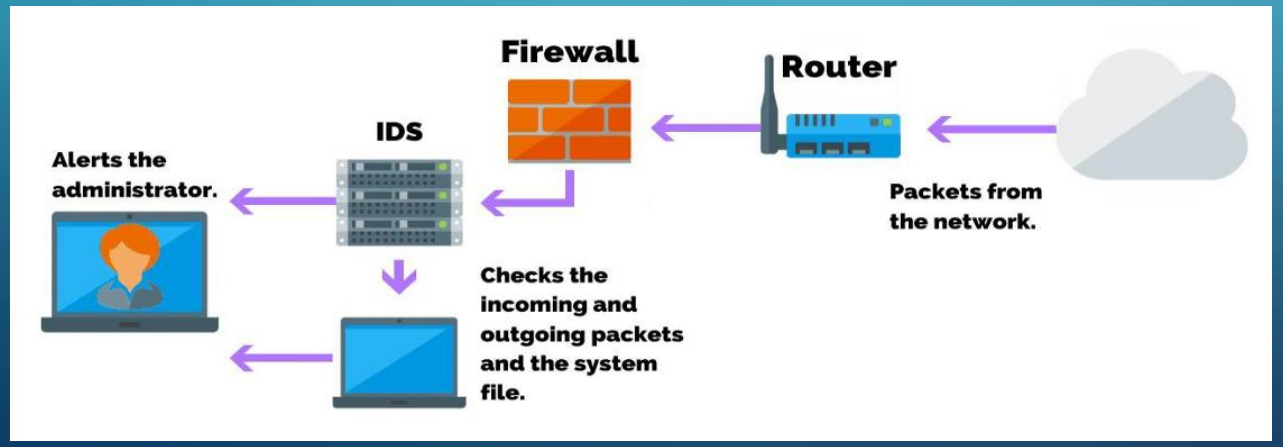
TYPES OF INTRUSION DETECTION SYSTEMS (CONTINUES…)

2. Host Intrusion Detection System (HIDS)

- Organizations install a HIDS on independent networked devices.
- HIDS examines the incoming and outgoing traffic of the **device only**.
- It detects suspicious activities on the device and alerts the administrator.
- its network card operates in **no promiscuous mode**.
- No promiscuous mode of operation can be an advantage in some cases, because not all NICs are capable of promiscuous mode.
- Promiscuous mode can be CPU intensive for a **slow** host machine.

TYPES OF INTRUSION DETECTION SYSTEMS (CONTINUES…)

- Many host-based IDSs **depend on signature** or string matching to detect threats, and they can be defeated by simply changing a tool enough that the signature no longer matches.
- Host-based systems were the first type of IDS to be developed and implemented.
- These programs are highly **effective for detecting insider abuses**.



TYPES OF INTRUSION DETECTION SYSTEMS (CONTINUES…)

3. Protocol-based Intrusion Detection System (PIDS)

- Organizations set up a PIDS at the front end of the server.
- It interprets the protocols between the server and the user.
- PIDS monitors the HTTPS server regularly to secure the web.

TYPES OF INTRUSION DETECTION SYSTEMS (CONTINUES…)

4. Application Protocol-based IDS (APIDS)

- Similarly, APIDS is set up within a group of servers.
- It interprets communication with the applications within the server to detect the intrusion.

TYPES OF INTRUSION DETECTION SYSTEMS (CONTINUES…)

5. Hybrid Intrusion Detection System

- is a **mixture** of two different IDS (**host** agents with **network** information).
- Hybrid System is **more responsive** and **effective** as compared to other IDS.

INTRUSION PREVENTION SYSTEMS (IPS)

UNIT 2

INTRUSION PREVENTION SYSTEMS

- IPS is a form of network security that works to **detect and prevent** identified threats.
- The IPS reports these events to system administrators and takes preventative action, such as closing access points and configuring firewalls to prevent future attacks.

HOW DO INTRUSION PREVENTION SYSTEMS WORK?

- Intrusion prevention systems work by scanning all network traffic.
- The IPS performs **real-time** packet inspection, deeply inspecting every packet that travels across the network.
- If any malicious or suspicious packets are detected, the IPS will carry out one of the following actions:

i. Terminate the TCP session

ii. Reconfigure the firewall

iii. Remove or replace any malicious content

HOW DO INTRUSION PREVENTION SYSTEMS DETECT MALICIOUS ACTIVITY?

1. **Signature-based Detection**

- Involves a dictionary of uniquely identifiable signatures located in the code of each exploit.

i. **Exploit-facing** – detect malicious activity based on common attack patterns.

ii. **Vulnerability-facing** – attempt to detect malicious activity by identifying specific vulnerabilities.

HOW DO INTRUSION PREVENTION SYSTEMS DETECT MALICIOUS ACTIVITY? (CONTINUES…)

2. Statistical Anomaly-based Detection

- Detection randomly sample network traffic and then compare the samples to a predetermined baseline performance level.

TYPES OF PREVENTION

1. **Signature-Based** – uses predefined signatures of well-known network threats.
2. **Anomaly-Based** – monitors for any abnormal or unexpected behavior on the network.
3. **Policy-Based** – This approach requires administrators to configure security policies according to organizational security policies and the network infrastructure.

TYPES OF INTRUSION PREVENTION SYSTEM (IPS)

1. **Network-based Intrusion Prevention System (NIPS)** – This type of intrusion prevention system has the ability to **monitor the whole network** and look for suspicious traffic by reviewing protocol activity.
2. **Wireless Intrusion Prevention Systems (WIPS)** – Only monitor wireless networks for suspicious activity by reviewing wireless networking protocols.
3. **Network Behaviour Analysis (NBA)** – Looks at network traffic in an effort to locate threats that cause unusual traffic flows, including distributed denial of service (DDoS) attacks and policy violations.
4. **Host-based Intrusion Prevention System (HIPS)** – Is an installed software package that looks into suspicious activity that occurs within a single host.

ANATOMY OF INTRUSION ANALYSIS

1. **Pre-Processing –** The data is collected about the intrusions, vulnerabilities and attacks and then it is putted down into classification scheme or pattern descriptors.
2. **Analysis –** Once the processing is completed, the analysis stage begins. The analysis engine looks for defined patterns that are known as attacks.
3. **Response –** The inline sensor can provide real time prevention through an automated response.
4. **Refinement –** This gives the opportunity to reduce false-positive levels and to have a more accurate security tool.

WHAT IS A VULNERABILITY ASSESSMENT?

- A **vulnerability assessment** is the process of defining, identifying, classifying and prioritizing vulnerabilities in computer systems, applications and network infrastructures.
- The process is sometimes referred to as vulnerability assessment/**penetration testing**, or **VAPT**.
- Importance of vulnerability assessments include –

i. Provides an organization with details on any **security weaknesses (Vulnerability Identification)**

ii. **Assess the risks** associated with those weaknesses **(Risk Assessment).**

iii. **Strengthening the Security Posture**

TYPES OF VULNERABILITY ASSESSMENTS

1. **Network-based scans** – are used to identify possible network security attacks.
2. **Host-based scans** – are used to locate and identify vulnerabilities in servers, workstations or other network hosts.
3. **Application scans** – test websites to detect known software vulnerabilities and incorrect configurations in network or web applications.

CREDENTIAL ANALYSIS NON CREDENTIAL ANALYSIS

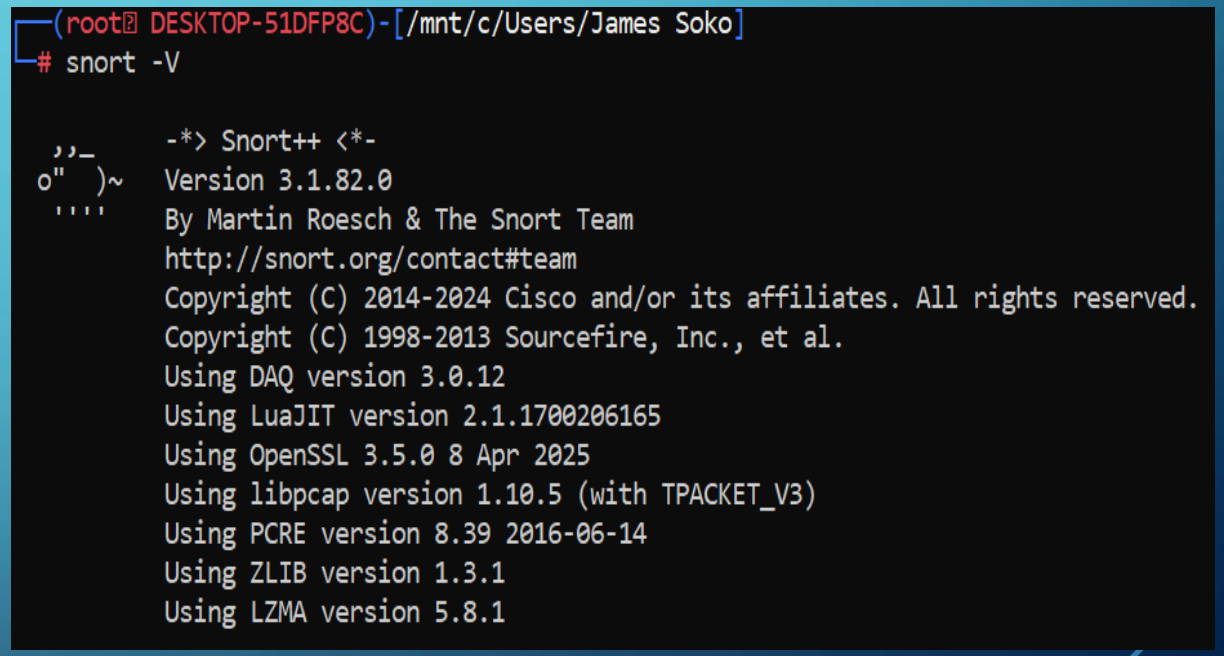
1. **Credential-based Vulnerability Assessments** – which make use of the admin account, do a more thorough check by looking for problems that cannot be seen from the network.
2. **Non-credentialed Scans** – provide a quick view of vulnerabilities by only looking at network services exposed by the host.

SNORT

UNIT 3

INTRODUCTION TO SNORT

- **Snort** is an open-source network based Intrusion Detection System (IDS) which is written in C programming language.
- It was developed in 1998 by Martin Roesch.
- Now it is developed by Cisco.



SNORT FEATURES

- It is based on library packet capture tool.
- Its features include:

1. **Real-time traffic monitor** – used to monitor the traffic that goes in and out of a network.
2. **Content matching** – collates rules by the protocol.
3. Creates logs
4. Default and Local Rules

SETTING UP YOUR HOME NETWORK

- Identify your active network interface.

```
┌──(root DESKTOP-51DFP8C)-[/etc/snort]
└─# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:17:5f:5b brd ff:ff:ff:ff:ff:ff
    inet 172.24.20.179/20 brd 172.24.31.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe17:5f5b/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```

- Alternatively, you can use **ifconfig** and **ip addr**.

SETTING UP YOUR HOME NETWORK (CONTINUES…)

- Edit the **snort.lua** or **snort.conf** file using **nano** command.

From

```
--------------------------------------------------
-- 1. configure defaults
--------------------------------------------------

-- HOME_NET and EXTERNAL_NET must be set now
-- setup the network addresses you are protecting
HOME_NET = 'any'
```

To

```
--------------------------------------------------
-- 1. configure defaults
--------------------------------------------------

-- HOME_NET and EXTERNAL_NET must be set now
-- setup the network addresses you are protecting
HOME_NET = '172.24.20.0/20'
```

PACKET SNIFFING MODE CONFIGURATIONS

- Edit the snort configuration file with **nano snort.lua** or **nano snort.conf**

From

```
ips =
{
    -- use this to enable decoder and inspector alerts
    -- enable_builtin_rules = true,

    -- use include for rules files; be sure to set your path
    -- note that rules files can include other rules files
    -- (see also related path vars at the top of snort_defaults.lua)

    variables = default_variables,
}
```

To

```
ips =
{
    -- use this to enable decoder and inspector alerts
    enable_builtin_rules = true,

    -- use include for rules files; be sure to set your path
    -- note that rules files can include other rules files
    -- (see also related path vars at the top of snort_defaults.lua)

    variables = default_variables,
}
```

SNORT COMMAND LINE OPTIONS

- Run Snort in **packet sniffing mode** using the configuration file.

```
┌──(root DESKTOP-51DFP8C)-[/mnt/c/Users/James Soko]
└─# cd /etc/snort/

┌──(root DESKTOP-51DFP8C)-[/etc/snort]
└─# snort -c snort.lua -i eth0
```

i. **-c** allows you to specify which configuration file you want to use. In snort v3 **snort.lua** is used while in v2 **snort.conf** is used.

ii. **-i** specifies which interface Snort should listen on e.g., Ethernet (**eth0**)

SNORT COMMAND LINE OPTIONS (CONTINUES…)

iii. **-V** displays the Snort **version** number and then exits.

iv. **-?** lists all switches and **options** and then exits.

```
┌──(root DESKTOP-51DFP8C)-[/etc/snort/rules]
└─# snort -?
-? <option prefix> output matching command line option quick help (same as --help-options) (optional)
-A <mode> set alert mode: none, cmg, or alert_*
-B <mask> obfuscated IP addresses in alerts and packet dumps using CIDR mask
-C print out payloads with character data only (no hex)
-c <conf> use this configuration
-D run Snort in background (daemon) mode
-d dump the Application Layer
```

SNORT COMMAND LINE OPTIONS (CONTINUES…)

- Snort itself uses something called **Data Acquisition library (DAQ)** to make abstract calls to packet capture libraries.

❑ Run Snort in inline mode (ensure interface and bridge setup if required)

```
┌──(root DESKTOP-51DFP8C)-[/etc/snort]
└─# snort --daq afpacket -c snort.lua -i eth0:lo -Q
```

- **--daq afpacket** tells Snort to use the **AFPacket DAQ module.**

v. **-Q** enables **inline mode** (for IPS).

SNORT ALERT MODES

- You can view these snort alert modes via nano **/etc/snort/snort.lua** in v3 and **/etc/snort/snort.conf** in v2.

```
--------------------------------------------------------------------------
-- 7. configure outputs
--------------------------------------------------------------------------

-- event logging
-- you can enable with defaults from the command line with -A <alert_type>
-- uncomment below to set non-default configs
--alert_csv = { }
  alert_fast = { }
--alert_full = { }
--alert_sfsocket = { }
--alert_syslog = { }
--unified2 = { }
```

SNORT ALERT MODES & SNORT COMMAND LINE OPTIONS (CONTINUES…)

1. **Fast**

```
┌──(root㉿ DESKTOP-51DFP8C)-[/etc/snort]
└─# snort -c snort.lua -i eth0 -A fast
```

vi. **-A** generates an alert using one of the specified alert-modes: **fast, full, none,** and **unsock**.

- Snort alerts report the **timestamp,** send an **alert message,** show the source and destination **IP address** and **port**.

SNORT ALERT MODES (CONTINUES…)

2. **Full**

- Is the **default alert mode**.
- Additionally to the information printed in the fast mode, the full mode shows the **TTL**, packet headers and datagram **length**, **service**, ICMP type, window size, **ACK** and sequence number.

3. **Console** – prints fast alerts in the console.

SNORT ALERT MODES (CONTINUES…)

4. **None** – With this mode, Snort **does not generate alerts.**
5. **Unsock** – This is useful to **export alert** reports to other programs through **Unix sockets.**
6. **Cmg** – This alerts mode was developed by Snort for **testing purposes;** it prints a full alert on the console without saving logs.

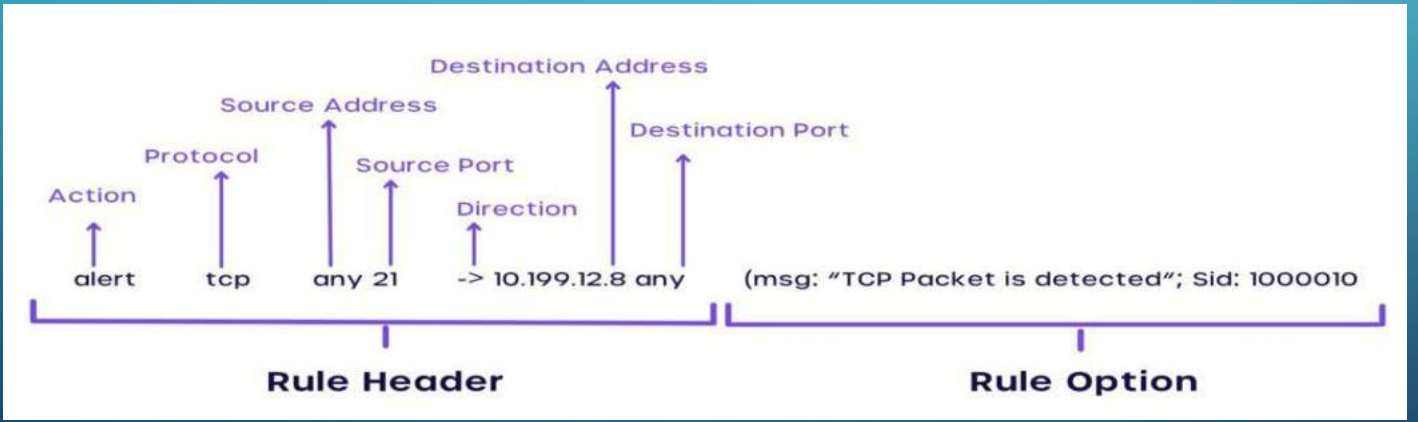
SNORT RULES

UNIT 4

WORKING WITH SNORT RULES

- Provide a sample Snort rule to detect HTTP traffic on port 80 and explain its structure.

- Navigate to nano **/etc/snort/rules/local.rules**



RULE HEADER

1. **Action**

i. **alert** – **alerts the network administrator** upon detecting a potential threat.

ii. **drop** – used in IPS (Intrusion Prevention System) mode to **block malicious traffic**.

iii. **pass** – ignores the packet.

iv. **log** – **records packet data** (headers and payload) but does not alert.

v. **activate** – generates an alert and then **activates another rule** (a dynamic rule).

vi. **dynamic** – does nothing until activated by an activate rule.

vii. **reject** – **drops the packet** and sends a TCP reset (for TCP) or ICMP unreachable (for UDP/ICMP).

viii. **sdrop (Silent Drop)** – silently drops the packet without alerting or logging.

RULE HEADER (CONTINUES…)

2. **Protocol –** you may key in the Protocol address that you need e.g., **tcp**.
3. **Source IP Address –** For example, **172.24.20.0** or **any**
4. **Source Port –** computers generally have 65,536 TCP Ports (e.g., **80** for HTTP) and if you wish, you may again key-in '**any**' to define all such ports in the rule.
5. **Flow –** refers to the **direction** of the traffic and denoted by **->**
6. **Destination IP Address**
7. **Destination Port**

WORKING WITH SNORT RULES (CONTINUES…)

- Below is our local rule:

```
  GNU nano 8.4                                          local.rules *
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# ----------------
# LOCAL RULES
# ----------------
# This file intentionally does not come with signatures.  Put your local
# additions here.
alert tcp any any -> any 80 (msg:"HTTP Traffic  Detected"; sid:1000001; rev:1;)
```

i. **msg** is the message displayed after the detection.

ii. **sid** is the snort ID (below 1 million for default rules and above that are local rules)

iii. **rev** is the revision number.

SNORT RULES EXAMPLES AND CHEAT SHEET

❑ **NOTE:**

1. You can replace **any** with an IP Address e.g., **172.24.20.0/24** or port e.g., **21**
2. You can also replace the IP Address **172.24.20.0/24** with **$HOME_NET**
3. You can also replace any with **$EXTERNAL_NET** if specified in the configuration file.

CONFIGURING LOCAL RULES

- Edit the snort configuration file and include the local rule path.

```
ips =
{
    -- use this to enable decoder and inspector alerts
    enable_builtin_rules = true,

    -- use include for rules files; be sure to set your path
    -- note that rules files can include other rules files
    -- (see also related path vars at the top of snort_defaults.lua)

    variables = default_variables,

    include = {
                RULE_PATH .. '/local.rules'
    }
}
```

SNORT COMMAND LINE OPTIONS (CONTINUES…)

- Test your snort configurations for any errors.

```
┌──(root DESKTOP-51DFP8C)-[/etc/snort]
└─# snort -c snort.lua -T
```

```
Snort successfully validated the configuration (with 0 warnings).
o")~   Snort exiting
```

vii.-T starts Snort in self-test mode. Useful for **debugging** Snort before it is run in daemon mode or before it is launched on a production box. Can be used for **testing the correctness** of your configuration files.

THE SNORT CONFIGURATION FILE

- A sample configuration file **snort.conf** is included in the Snort distribution.
- You use the **-c** command line switch to specify the name of the configuration file.
- This file contains six basic sections:

1. **Variable definitions** – where you define different variables.
2. **Configure parameters** – specify different Snort configuration options.

THE SNORT CONFIGURATION FILE (CONTINUES…)

3. **Preprocessor configuration** – are used to perform certain actions before a packet is operated by the main Snort detection engine.
4. **Output module configuration** – control how Snort data will be logged.
5. **Defining new action types**
6. **Rules configuration and include files** – These files are then included inside the main configuration file using the **include** keyword.

ANALYSIS CONSOLE FOR INTRUSION DATABASES (ACID)

UNIT 5

NOTICE

- The slides will be added before Monday, for now study the lecture slides.

ACCESS CONTROL

CHAPTER TWO

WHAT IS ACCESS CONTROL?

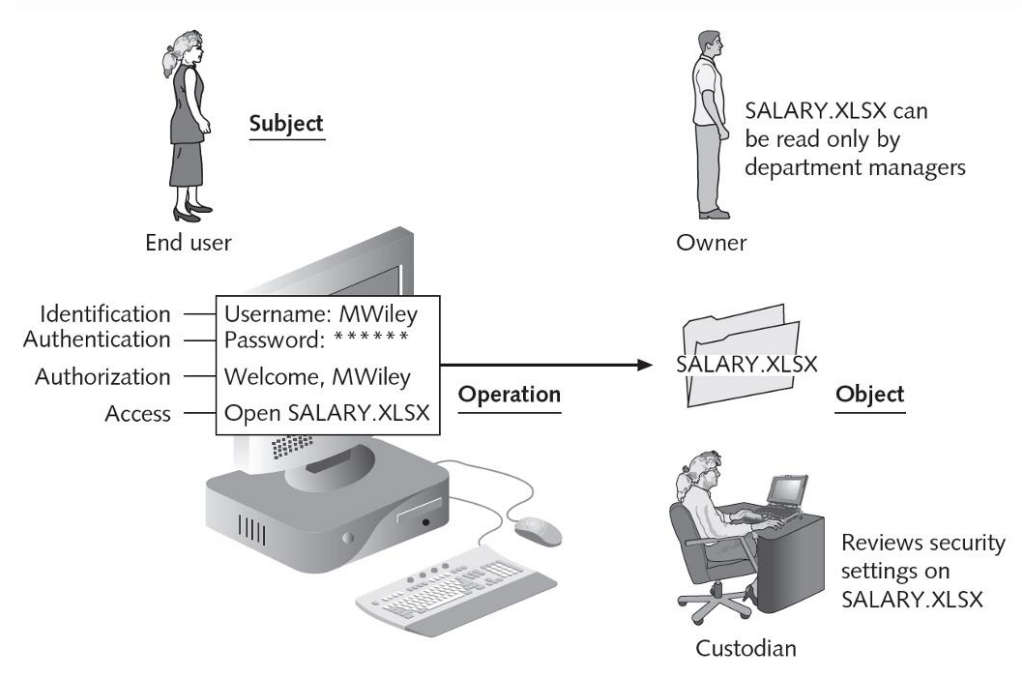
- Granting or denying approval to use specific resources.

1. **Identification** – Presenting credentials.
2. **Authentication** – Checking the credentials.
3. **Authorization** – Granting permission to take action.
4. **Access** – Right given to access specific resources.

ACCESS CONTROL TERMINOLOGY

5. **Object** – Specific resource (e.g., file)
6. **Subject** – User or process functioning on behalf of a user (i.e., computer user)
7. **Operation** – Action taken by the subject over an object (e.g., deleting a file)

ACCESS CONTROL TERMINOLOGY (CONTINUES…)



8. **Owner** – Determines the level of security needed to the data.
9. **Custodian** – Assigned by the owner to review security settings.
10. **End User** – Accesses the information and does not circumvent security based on the organizations guidelines.

ACCESS CONTROL MODELS

1. **Mandatory Access Control (MAC)**

- Most restrictive access control model.
- Typically found in military settings.
- It has two elements –

i. **Labels** – indicate level of privilege

ii. Levels

- MAC grants permissions by matching object labels with subject labels.
- To determine if file may be opened:

i. Compare object and subject labels

ii. Subject must have equal or greater level than object to be granted access

ACCESS CONTROL MODELS (CONTINUES…)

- Two major implementations of MAC

i. **Lattice model –**

ii. **Bell-LaPadula** – Subjects may not create a new object or perform specific functions on lower level objects.

ACCESS CONTROL MODELS (CONTINUES…)

2. Discretionary Access Control (DAC)

- Least restrictive model.
- Every object has an owner.
- Owners have total control over their objects.
- Owners can give permissions to other subjects over their objects.

ACCESS CONTROL MODELS (CONTINUES…)

3. **Role Based Access Control (RBAC)**

- Also called **Non-discretionary Access Control**.
- Access permissions are based on user's job function.

4. **Rule Based Access Control (RBAC)**

- Dynamically assigns roles to subjects based on a set of rules defined by a custodian.

BEST PRACTICES FOR ACCESS CONTROL

1. **Separation of duties** – Fraud can result from single user being trusted with complete control of a process.
2. **Job rotation** – Limits amount of time individuals are in a position to manipulate security configurations.
3. **Least privilege** – Helps reduce attack surface by eliminating unnecessary privileges.
4. **Implicit deny** – If a condition is not explicitly met, access request is rejected.

ACCESS CONTROL LISTS

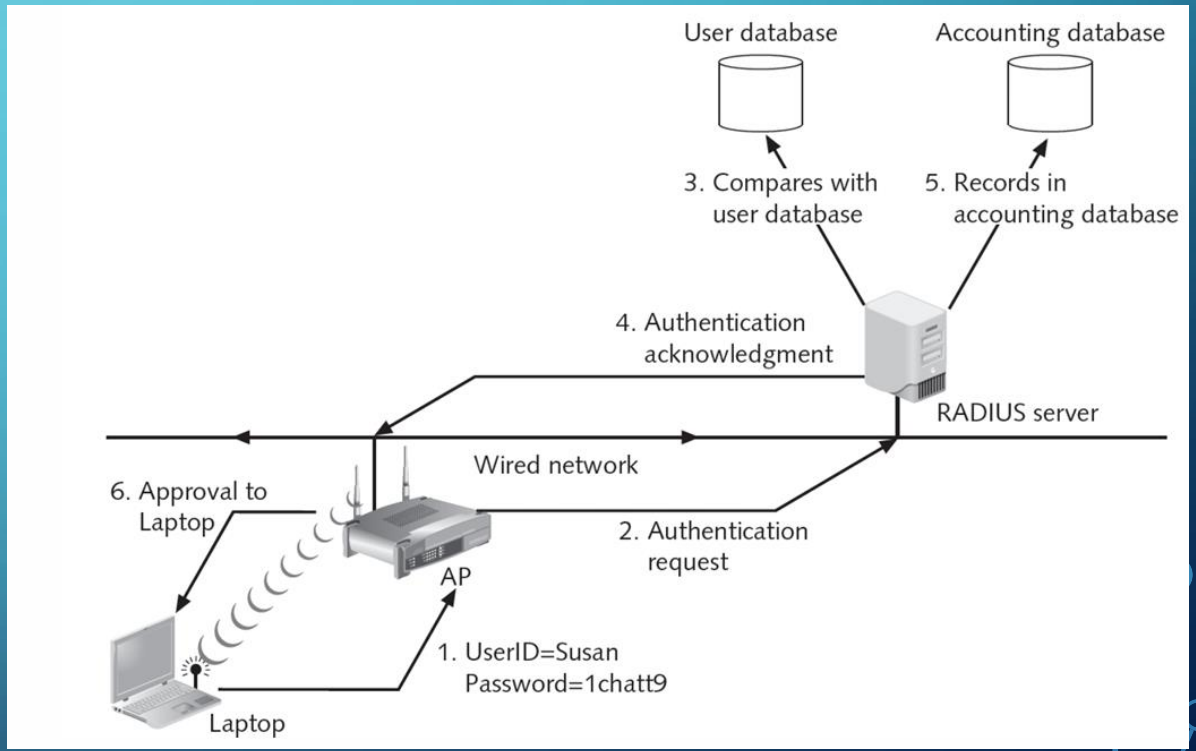
- Set of permissions attached to an object.
- When subject requests to perform an operation, system checks ACL for an approved entry.
- Each entry in the ACL table is called Access Control Entry (ACE).

AUTHENTICATION SERVICES

- **Authentication** is process of verifying credentials.

1. **Remote Authentication Dial-In User Service (RADIUS)**

- Developed in 1992
- **RADIUS client** are devices such as a Wireless AP responsible for sending user credentials and connection parameters to the RADIUS server.



TERMINAL ACCESS CONTROLLER ACCESS CONTROL SYSTEM PLUS (TACACS+)

2. Terminal Access Controller Access Control System Plus (TACACS+)

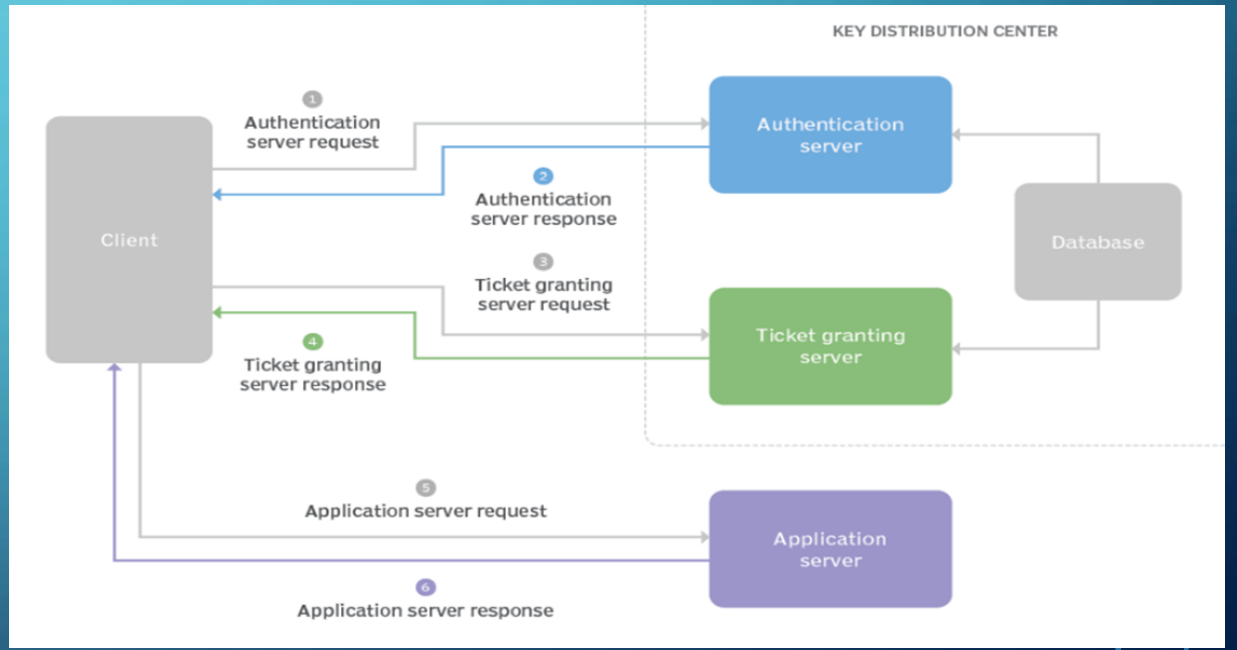
- Is an authentication service similar to RADIUS which was developed by **Cisco** Systems.

Feature	RADIUS	TACACS+
Protocol	UDP	TCP
Transport Characteristics	Connectionless, faster but less reliable	Connection-oriented, more reliable
Encryption	Encrypts the password in the packet	Encrypts the entire packet payload
Authentication & Authorization	Combined Processes	Separate Processes
Typical use	Remote user network access authentication	Device administration (routers, switches)

AUTHENTICATION SERVICES (CONTINUES…)

3. Kerberos

- Authentication system developed at MIT.
- Uses **encryption** and **authentication** for security.
- **Kerberos** is a protocol for authenticating service requests between trusted hosts across an untrusted network, such as the internet.



AUTHENTICATION SERVICES (CONTINUES…)

4. **Lightweight Directory Access Protocol (LDAP)**

- Keeps track of network resources and user's privileges to those resources.
- **LDAP** is a simpler subset of DAP which was designed to run over TCP/IP.
- Grants or denies access based on its information.
- **X.500** standard defines protocol for client application to access the DAP.
- **Weakness of LDAP** can be subject to **LDAP injection attacks.**